CHATHUKA DILAKSHANA

SOFTWARE ENGINEER | FULL STACK DEVELOPER | MERN STACK DEVELOPER

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Professional Summary

Software Engineering undergraduate with hands-on experience in Full-Stack Development (Laravel, MERN Stack) and ERP Systems. Proven ability to build scalable web applications, optimize business workflows, and develop AI-powered solutions using Machine Learning and Deep Learning. Seeking a Software Engineer role to contribute strong problem-solving and development skills.

Technical Skills

- Programming Languages: PHP, Java, JavaScript, Python, SQL, HTML
- Web Frameworks & Libraries: Laravel, MERN Stack (MongoDB, Express.js, React.js, Node.js), Tailwind CSS, Bootstrap 5
- Databases & Data Management: MySQL, MongoDB, Supabase, Data Integrity, CRUD Operations
- Tools & Version Control: Git/GitHub, Visual Studio, Postman, Notion, Jupyter Notebook
- Specialized Domains: ERP Systems Development, RESTful APIs, UI/UX Design, Machine Learning (ML), Deep Learning (DL), Full-Stack Web Development

Work Experience

Apps Technologies (Pvt) Ltd | Software Developer Intern **Jan 2025 - Aug 2025**

- Designed and deployed 3 mission-critical ERP modules (Inventory Report, Requisitions, Finance workflows) utilized by 5+ internal teams using Laravel, Bootstrap 5, and JavaScript.
- Engineered robust backend architecture, including transaction reference codes and auto-generated account structures, improving overall data integrity by an estimated 99.8%.
- Troubleshooted and resolved over 35 critical production bugs across core logistics modules (Stock Issue, Stock Transfer, and POS), optimizing reliability and increasing system uptime by 15%.
- Performed comprehensive QA and regression testing, identifying edge cases and streamlining deployment workflows to enable smooth, low-risk system updates.
- Delivered responsive UIs featuring dynamic dropdowns, pagination, and export functions, streamlining reporting efficiency for 50+ end-users.
- Collaborated cross-functionally with QA teams and stakeholders, producing technical documentation and conducting live demonstrations.

Project Experience

AI-Powered Sinhala Sign Language Platform - Python, Flask, React.js, NLP | Jul 2025 - May 2026

- **Developed** a React.js web interface and a custom NLP engine to tokenize Sinhala text and map vocabulary to standardized semantic IDs.
- **Engineered** a rule-based algorithm to restructure standard Sinhala grammar into the visual-spatial Time-Subject-Object-Verb (TSOV) format.
- **Integrated** a Word2Vec machine learning model to accurately identify and process unrecognized slang words via intelligent semantic matching.
- **Optimized** real-time video generation using MoviePy, stitching and streaming multimodal 60 FPS animations in under 200 milliseconds

Deep Learning Cancer Data Analysis - CNN, TensorFlow, Keras | Dec 2025

- Benchmarked four Deep Learning architectures (including ResNet-DNN and Wide & Deep Network) for breast cancer classification on the Wisconsin Diagnostic Dataset.
- Attained 98.8% classification accuracy and successfully achieved the critical clinical goal of zero false negatives (1.00 Malignant Recall).

ViralFlow AI (SaaS Platform) | Next.js, Node.js, MongoDB, Tailwind CSS | Apr 2026

- **Architected** and launched an MVP SaaS platform for automated video transcription and subtitle generation to streamline social media content creation workflows.
- **Developed** a scalable full-stack application leveraging Next.js App Router for the frontend and Node.js for backend processing.
- **Integrated** AI-driven processing pipelines to deliver highly accurate text generation and seamless user experiences.
- **Deployed** the application to production, ensuring scalability and robust performance **(Live:** tryviralflow.vercel.app**).**

IoT Intrusion Detection System - Python, XGBoost, Machine Learning | Apr 2026

- **Engineered** a robust data pipeline to preprocess the massive CICIoT2023 dataset using RobustScaler and physical data balancing techniques.
- **Optimized** an XGBoost model via GPU-accelerated training to efficiently classify over 1.5 million records across 33 distinct cyber-attack categories.
- **Achieved** a 99.37% overall accuracy and an 82.00% Macro F1-Score on unseen test data, successfully identifying rare application-layer threats.

Community Empowerment Hub – MERN Stack, OpenAI API | May 2025

- **Developed** a comprehensive platform integrating a marketplace and job portal with role-based access control for Sellers, Buyers, and Admins.
- **Implemented** secure authentication mechanisms using JWT to protect user data and ensure secure transactions.
- **Integrated** OpenAI APIs to provide specialized AI-powered tools that enhance platform utility and user interaction.

Ceylon Buddy – Tour Guide Web Application | Apr 2025

- **Designed** and developed a fully responsive tour guide web platform for a client, helping travellers explore Sri Lankan destinations and attractions.
- **Deployed** and hosted the website **live at** ceylonbuddytour.com, ensuring scalability, security, and optimal performance.

Review Analysis Project – Python (Jupyter Notebook) | Jul 2024

- **Built** an NLP preprocessing pipeline for a dataset of 10,000+ customer reviews utilizing CountVectorizer and TF-IDF.
- **Trained** and evaluated Logistic Regression and Naive Bayes models to classify sentiments, achieving an average accuracy of 88%.

Tourist Management System – MERN Stack | Dec 2024

- **Developed** a full-stack web application to manage tourism services, facilitating seamless CRUD operations for bookings and travel packages.
- **Engineered** secure authorization protocols to effectively manage and protect customer and administrative data.

Education

Sri Lanka Institute of Information Technology (SLIIT) 2022 - Present (Expected 2026)
BSc (Hons) in Information Technology, Specializing in Information Technology (Undergraduate)

ST/ALOYSIUS COLLEGE Rathnapura 2021/2022
G.C.E Advanced Level - Maths Stream

References

Mr. Buddika Harshanath — Senior Lecturer, Dept. of Information Technology, SLIIT | harshanath.s@sliit.lk
Mrs. Dushanthi Sadeepa Kuruppu — Lecturer (Tenure-Track), Faculty of Computing, SLIIT | dushanthi.k@sliit.lk